EXPRESS.JS FRAMEWORK

Theory Assignment:

Introduction:

What is express.js?

Express.js is a fast, minimal and flexible web framework for node.js

It helps developers build web servers and APIs easily

Advantages of express.js over the built-in http module

Express.js : easy to use, less code, built-in routing, middleware support, easy handling of requests/responses, faster development

http module: more complex, more code, manual routing, no direct middleware support, manual handling required, slower development

main advantages of express.js

1) Simple routing

Express makes route creation easy:

Ex: app.get("/about", (req, res)=>{

Res.send("about page); });

2) Middleware support

Express supports middleware function for:

Loggin, authentication, validation, error handling

Ex: app.use((req, res, next) =>{

Console.log(req,method);

Nex();});

3) Cleaner code
   Express reduce boilerplate code and improves readability.

4) Easy handling of requests & responses
   Express provides:
   Req.params
   Req.query
   Req.body
   This makes data handling easier

5) Rest API development
   Express is widely used for creating restful apis quickly

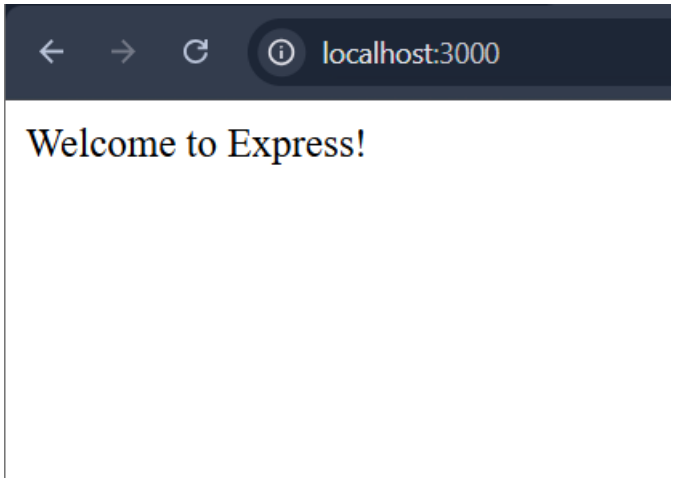
Practical assignment:

1) Setting up a basic express server
   Create project folder
   Mkdir express-app
   Cd express-app
   Npm init -y
   Npm install express

   Server.js

```
const express = require("express");

const app = express();

const PORT = 3000;

// Root Route
app.get("/", (req, res) => {
```

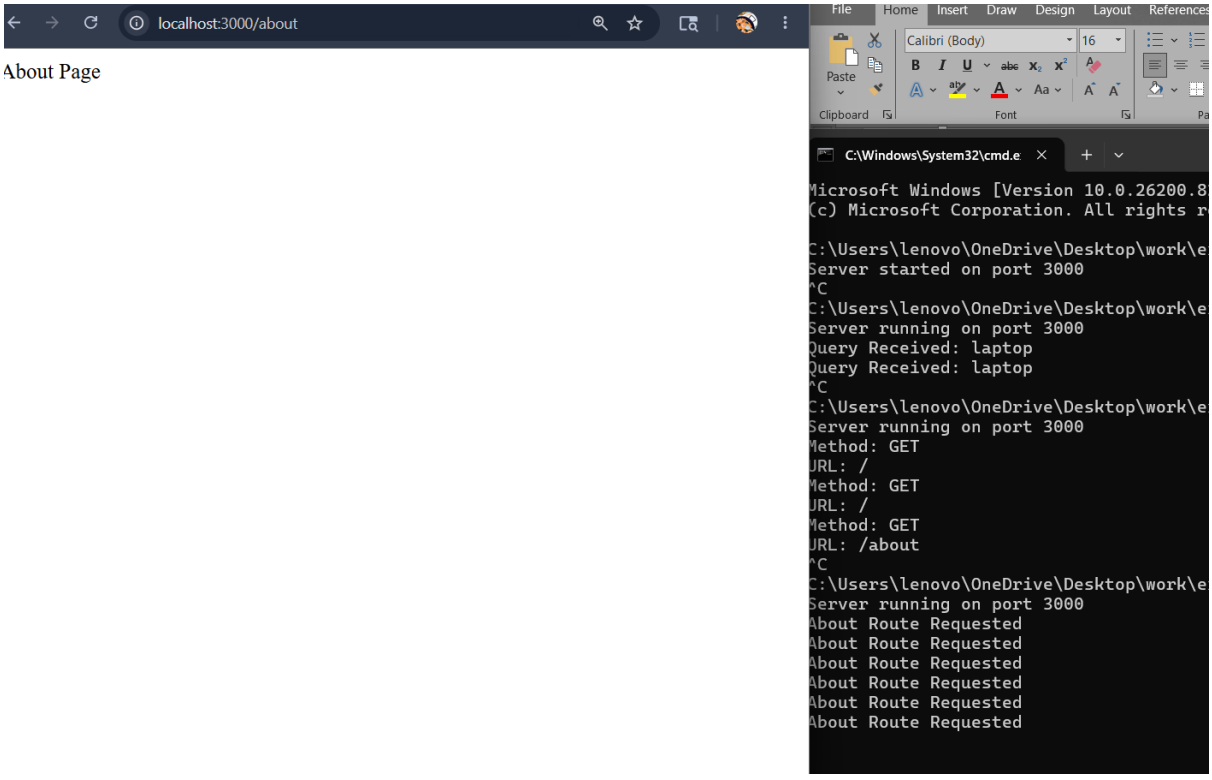
```
    res.send("Welcome to Express!");
});

// Server Start
app.listen(PORT, () => {
    console.log(`Server started on port ${PORT}`);
});
```



2) Creating routes: multiple routes

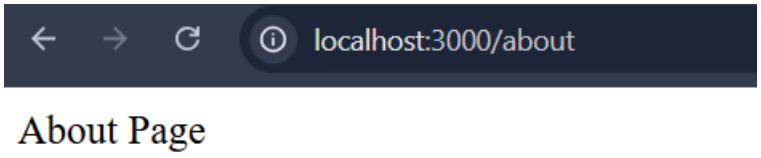
```
const express = require("express");

const app = express();

const PORT = 3000;

// Home Route
app.get("/", (req, res) => {
    console.log("Home Route Requested");
    res.send("Home Page");
});
```

```
// About Route
app.get("/about", (req, res) => {
   console.log("About Route Requested");
   res.send("About Page");
});

// Contact Route
app.get("/contact", (req, res) => {
   console.log("Contact Route Requested");
   res.send("Contact Page");
});

// Server Start
app.listen(PORT, () => {
   console.log(`Server running on port ${PORT}`);
});
```



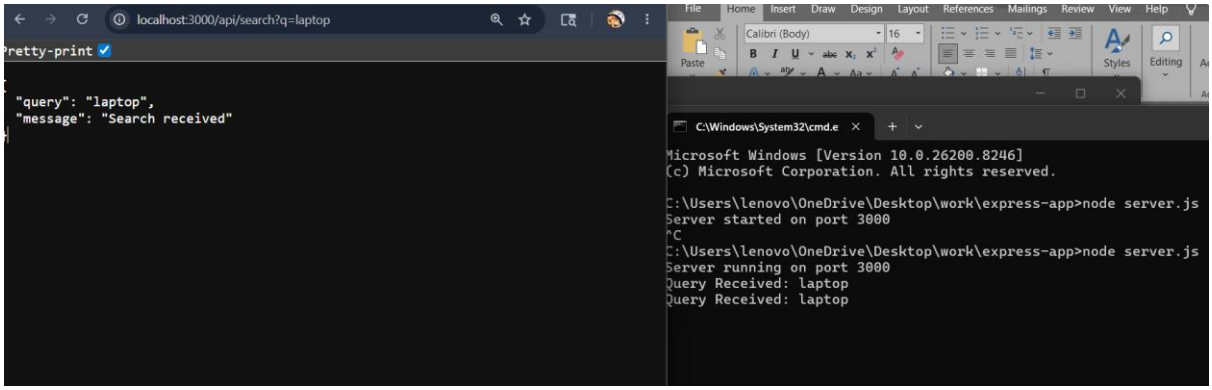
3) Using express middleware

```
const express = require("express");

const app = express();

const PORT = 3000;

// Custom Middleware
app.use((req, res, next) => {
   console.log(`Method: ${req.method}`);
   console.log(`URL: ${req.url}`);

   next(); // Important
});

// Routes
app.get("/", (req, res) => {
   res.send("Home Page");
});

app.get("/about", (req, res) => {
   res.send("About Page");
});

// Server Start
app.listen(PORT, () => {
   console.log(`Server running on port ${PORT}`);
});
```



4) Handling query parameters:

```
const express = require("express");

const app = express();

const PORT = 3000;

// Search Route
app.get("/api/search", (req, res) => {

   const searchTerm = req.query.q;

   console.log("Query Received:", searchTerm);

   res.json({
      query: searchTerm,
      message: "Search received"
   });
});

// Server Start
app.listen(PORT, () => {
   console.log(`Server running on port ${PORT}`);
});
```



5) Get, post, delete routes example:

```
const express = require("express");

const app = express();

const PORT = 3000;

// Middleware
app.use(express.json());

// GET Route
app.get("/", (req, res) => {
   res.send("GET Request");
});

// POST Route
app.post("/user", (req, res) => {
```

```
    console.log(req.body);

    res.json({
        message: "POST Request Successful",
        data: req.body
    });
});

// DELETE Route
app.delete("/delete/:id", (req, res) => {

    const userId = req.params.id;

    res.json({
        message: "Delete Request Successful",
        deletedId: userId
    });
});

// Server Start
app.listen(PORT, () => {
    console.log(`Server running on port ${PORT}`);
});
```

Important express methods

Method

App.get() => fatch data

App.post()=> insert data

App.put()=> update data

App.delete() => delete data

App.use() => middleware

res.send() => send response

Res.json()=> send json

Handling form data with post requests

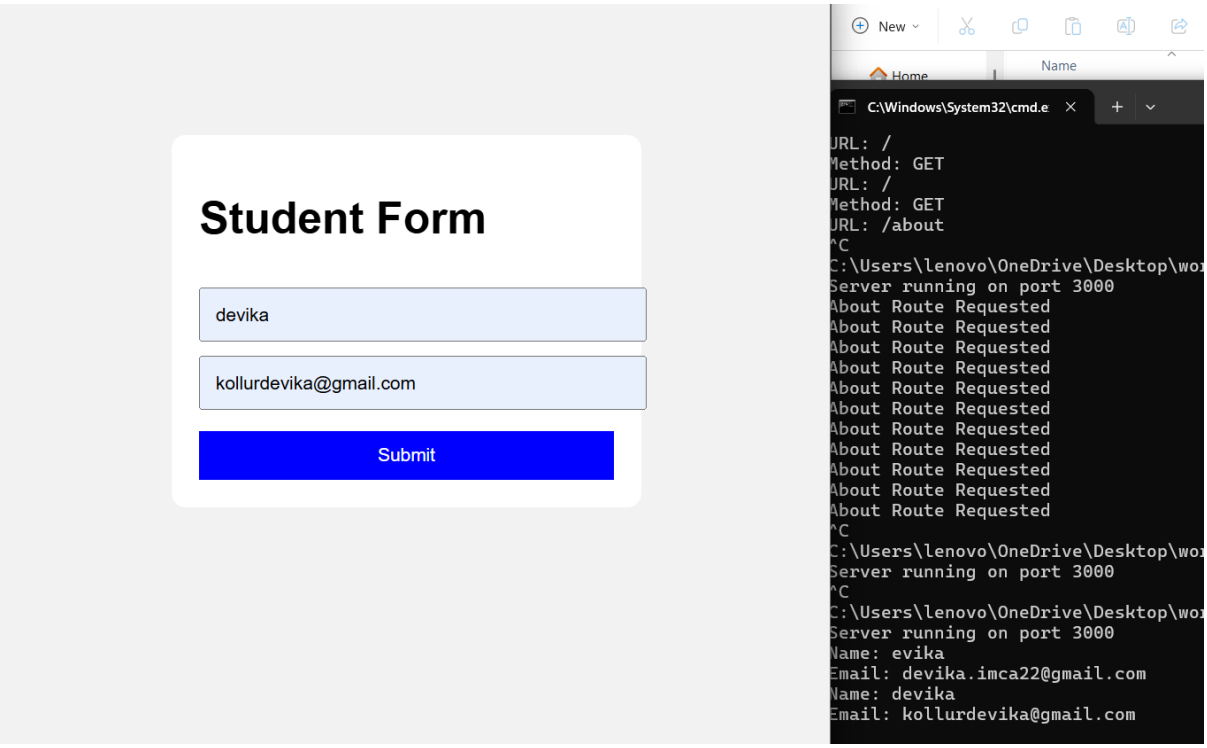
Server.js

```
const express = require("express");
const path = require("path");

const app = express();

const PORT = 3000;

// Middleware for form data
app.use(express.urlencoded({ extended: true }));

// Serve Static Files
app.use(express.static("public"));
```

```
// Root Route
app.get("/", (req, res) => {
   res.sendFile(path.join(__dirname, "public", "index.html"));
});

// POST Route for Form Submission
app.post("/submit", (req, res) => {

   const { name, email } = req.body;

   console.log("Name:", name);
   console.log("Email:", email);

   res.send(`
      <h1>Form Submitted Successfully</h1>
      <h2>Name: ${name}</h2>
      <h2>Email: ${email}</h2>
   `);
});

// Start Server
app.listen(PORT, () => {
   console.log(`Server running on port ${PORT}`);
```

});

Output:


Student Form
devika
kollurdevika@gmail.com
Submit
New
Name
Home
C:\Windows\System32\cmd.e
URL: /
Method: GET
URL: /
Method: GET
URL: /about
^C
C:\Users\lenovo\OneDrive\Desktop\wo
Server running on port 3000
About Route Requested
About Route Requested
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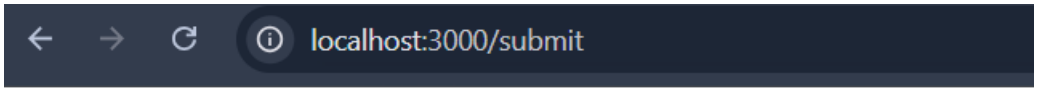
^C
C:\Users\lenovo\OneDrive\Desktop\wo
Server running on port 3000
^C
C:\Users\lenovo\OneDrive\Desktop\wo
Server running on port 3000
Name: evika
Email: devika.imca22@gmail.com
Name: devika
Email: kollurdevika@gmail.com



Form Submitted Successfully

Name: devika

Email: kollurdevika@gmail.com

Understanding important part:

1) Express.urlencoded()
   App.use(express.urlencoded({ extended: true}));
   Used to read form data from req.body

2) Serving static files

   App.use(express.static("public"));

   This servers:

   Html, css, javascript, images from the public folder

3) form post method
   <from action="/submit" method="post">

Action=”submit”=> sends data to /submit
Method=”post”=> send data securely

4) access from data
const {name, email } = req.body;

implementing error handling middlewre

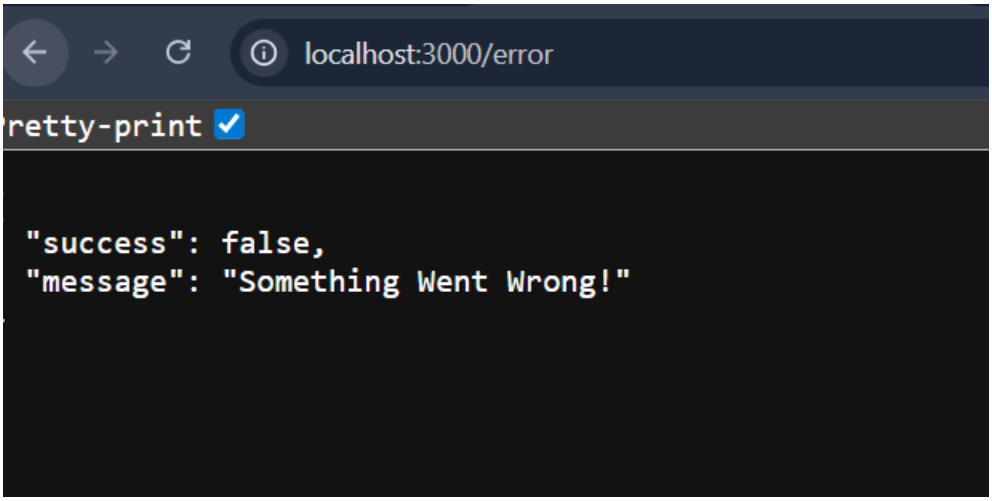
what is error handling middleware?
In express.js error handling middleware is used to:
Catch application errors
Prevent server crashes
Send proper error responses
(err, req, res, next)

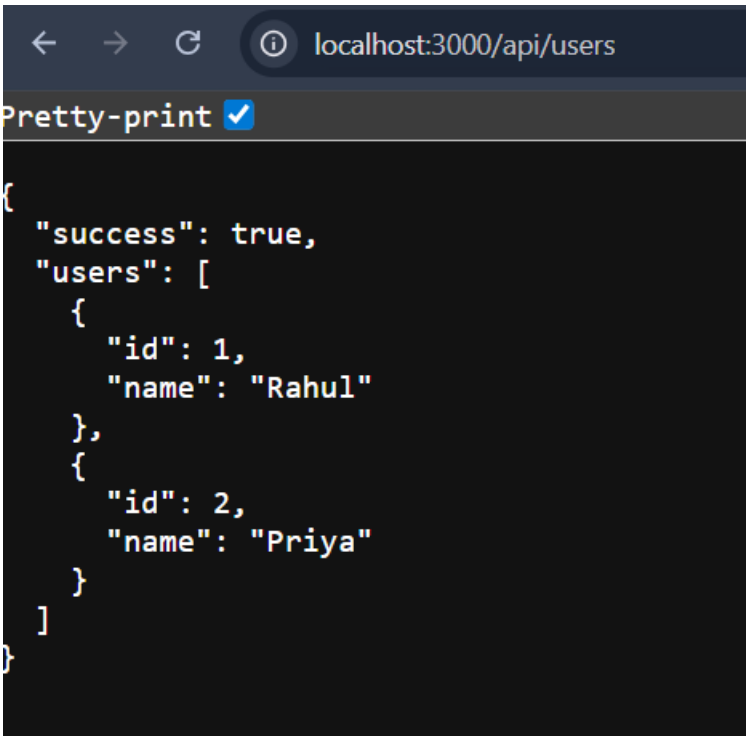
Server.js

```
const express = require("express");

const app = express();

const userRoutes = require("./routes/users");

const PORT = 3000;

// Middleware
app.use(express.json());


// Home Route
```

```
app.get("/", (req, res) => {
   res.send("Home Page");
});


// Route to Simulate Error
app.get("/error", (req, res, next) => {

   const error = new Error("Something Went Wrong!");

   next(error);
});


// User Routes
app.use("/api/users", userRoutes);



// Error Handling Middleware
app.use((err, req, res, next) => {

   console.log("Error Middleware Called");

   res.status(500).json({
      success: false,
      message: err.message
   });
});


// Server Start
app.listen(PORT, () => {
```

```
    console.log(`Server running on port ${PORT}`);
});
```





1) express router:
   const router = express.router();
   used to create moduler routes.
2) Import router

Const userRoutes = require(“./routes/users”);
Imports routes into main app

3) Use router
   App.use(“/api/users”, useRoutes);
   Base route: /api/users

4) Error middleware
   App.use((err, req, res, next)=> {
   Catches all application errors

5) Next(error)
   Next(error);
   Passes error to error middleware

   Why use router?

   Clean code
   Moduler structure
   Easy maintenance
   Better project organization